

ANUSHKA DEV.

Deep Learning & Computer Vision Engineer — Multimodal AI — Open to ML Intern / Junior ML Roles

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Portfolio • GitHub • LinkedIn

ABOUT ME

Deep Learning & Computer Vision engineer specializing in multimodal AI systems, real-time inference, and experiment-driven model optimization. Experienced in building production-ready pipelines including dataset creation, GPU-accelerated training, model evaluation, and deployment. Strong research mindset, fast learner, and highly motivated to contribute to ML Intern / Junior ML Engineering roles.

CORE SKILLS

Deep Learning: CNNs, Hybrid Multimodal Models, Transfer Learning, 3-Stage Progressive Training, Label Smoothing, LR Scheduling, Mixed Precision (FP16)

Computer Vision: MediaPipe Hands, Keypoint Extraction, Skeleton Rendering, OpenCV

Model Optimization: Early Stopping, One-Cycle LR, Checkpointing, Augmentation Pipelines

Evaluation: Confusion Matrix, Classification Report, ROC, Precision-Recall, Per-Class Analysis

Frameworks & Tools: PyTorch, TensorFlow, Streamlit, FastAPI, NumPy, Scikit-Learn, Git

Systems: CUDA, GPU Profiling, WSL2 Linux, Build/Deploy (basic)

Languages: Python

EXPERIENCE

Deep Learning Research Engineer — Independent Applied Research 2024 — Present

- Designed and trained multimodal deep learning models combining RGB images + skeleton keypoints.
- Built a full training pipeline with **progressive unfreezing**, FP16 mixed precision, One-Cycle LR, and per-stage checkpoints (fusion, skeleton, fine-tune).
- Managed **150K+ images**, performing dataset cleaning, augmentation, leakage checking, and class-balance validation.
- Implemented evaluation suite including confusion matrix, per-class accuracy, PR/ROC curves and classification reports.
- Integrated real-time inference using OpenCV + MediaPipe + Streamlit achieving **~100ms prediction latency**.

PROJECTS

Multimodal Sign Language Translator (Real-time) *PyTorch, MediaPipe, Streamlit, React*

- Developed a **Dual-Branch Multimodal Late-Fusion Model** combining EfficientNet-B0 (RGB) and CNN (skeleton).
- Implemented **3-Stage Transfer Learning** with progressive unfreezing achieving **97.9% validation accuracy**.
- Built real-time inference pipeline (frame capture → Mediapipe landmarks → skeleton rendering → multimodal prediction).
- Added stabilization, debouncing, and FP16 inference for fast and accurate predictions.
- Frontend prototype built using Streamlit; React + WebCam backend integration under development.
- **GitHub:** github.com/anuushka-dev/Sign-Language-translator

Deepfake Image Detection

TensorFlow, ResNet50, Streamlit

- Constructed a deepfake detection pipeline using **hybrid CNN+ResNet50** trained on 141K images.
- Built dataset cleaning workflow using perceptual hashing, augmentation, and stratified splitting.
- Implemented model monitoring with ROC/PR curves, confusion matrix, learning curve analysis.
- **GitHub:** github.com/anuushka-dev/Deepfake-Image-Detection

EDUCATION

B.Tech Computer Science & Engineering (AI/ML)

Dr. Ambedkar Institute of Technology for Divyangjan

2023 — 2027

ACHIEVEMENTS

- Achieved **97–98% accuracy** on multimodal ASL classifier.
- Built fully automated experiment tracking, dataset validation and evaluation suite.
- Delivered real-time inference pipeline with optimized PyTorch model.

INTERESTS

Multimodal AI • Computer Vision • Real-time Systems • Applied Research